

The renewable energy transition requires careful grid planning.

Solar and wind reduce emissions but need storage to balance supply.

Grid operators must manage variability through battery storage, demand response, and

Policy incentives accelerate deployment, yet land use and supply chains remain constraints.

Interconnection upgrades.

A just transition protects workers in fossil industries while building new clean energy jobs.

Modelling shows that combining renewables with firm low-carbon capacity improves reliability.

International cooperation on technology transfer helps developing economies leapfrog

carbon-intensive infrastructure.